

July 2025 CSE 220
Online on Signals and Their Properties
Subsections: B1, B2
Total Marks: 10
Total Time: 30 Minutes

January 7, 2026

In this assignment, you need to implement phase change and time shift in discrete-time sinusoidal signals. Then you need to demonstrate how time shift and phase change are related to each other in these signals. You shall make use of the provided Python template file.

Tasks

1. Generate the time axis n and compute the discrete-time sinusoid $x[n] = A \cos(\Omega_0 n + \phi)$ in the function `sinusoid(...)`. Refer to the template file for more details.
2. Implement the functions `time_shift_sinusoid(...)` and `phase_change_sinusoid(...)`.
3. We will use mean-squared-error to demonstrate the difference between the signals after time shift and after phase change (already implemented in the template file).
4. Plot appropriate graphs and demonstrate whether **time shifting** corresponds to **phase change** in discrete-time sinusoidal signal. You should be able to explain your findings to the instructor.
5. Plot appropriate graphs and demonstrate whether **phase change** corresponds to **time shifting** in discrete-time sinusoidal signal. You should be able to explain your findings to the instructor.

Marks Distribution

- Implementing $x[n]$: 1 Mark
- Implementing time shift: 2 Marks
- Implementing phase change: 2 Marks
- Demonstrating how time shift is related to phase change in $x[n]$: 2 Marks
- Demonstrating how phase change is related to time shift in $x[n]$: 3 Marks